Heat Transfer Analysis of Composite Rod with Insulation

1. Problem Description

Heat is supplied at one end of a cylindrical copper rod. The rod is covered with insulation, and heat is lost to the surroundings through the insulation by convection and radiation.

Heat flow occurs:

- Axially along the rod
- Radially through insulation
- From insulation surface to air

2. Boundary Conditions

(1) At the base ($x = 0$)

Heat input is known:

$$Q = VI = 5.2 \times 1.70 = 8.84\,W$$

$$-kA\frac{dT}{dx}\bigg|_{x=0} = Q$$

(2) At the rod-insulation interface ($r = r_1$)

$$T_{rod} = T_{ins}$$
$$k_{rod}\frac{\partial T}{\partial r} = k_{ins}\frac{\partial T}{\partial r}$$

(3) At the outer surface ($r = r_2$)

Heat loss occurs by convection and radiation:

$$-k_{ins}\frac{\partial T}{\partial r} = h_o(T - T_\infty) + \epsilon\sigma(T^4 - T_\infty^4)$$

(4) Ambient condition

$$T = T_\infty$$

(5) Steady state assumption

$$\frac{d}{dt} = 0$$

This implies:

$$\text{Heat in} = \text{Heat out}$$

3. Energy Balance Approach

Instead of solving the full 2D conduction equation, we use an energy balance over the rod.

$$Q = \int_0^L \frac{T(x) - T_\infty}{R'_{total}} dx$$

Let:

$$\theta = T - T_\infty$$

Then:

$$Q = \int_0^L \frac{\theta(x)}{R'_{total}} dx$$

Approximating using average temperature:

$$Q \approx \frac{\theta_{avg} \cdot L}{R'_{total}}$$

4. Temperature Data

Measured temperatures:

$$T = [157.7,\ 147.8,\ 133.5,\ 126.7,\ 124.3]$$

Ambient:

$$T_\infty = 31.3^\circ C$$

$$\theta = T - T_\infty = [126.4,\ 116.5,\ 102.2,\ 95.4,\ 93]$$

$$\theta_{avg} \approx 106$$

5. Total Resistance

$$R'_{total} = \frac{\theta_{avg} \cdot L}{Q}$$

$$R'_{total} = \frac{106 \times 0.117}{8.84} \approx 1.40 \text{ K·m/W}$$

6. Insulation Resistance

$$R'_{ins} = \frac{\ln(r_2/r_1)}{2\pi k_{ins}}$$

$$r_1 = 0.00705\,m, \quad r_2 = 0.02025\,m$$

$$R'_{ins} \approx 0.70 \text{ K·m/W}$$

7. Outer Resistance

$$R'_{out} = R'_{total} - R'_{ins}$$

$$R'_{out} = 1.40 - 0.70 = 0.70$$

8. Heat Transfer Coefficient

$$R'_{out} = \frac{1}{h_{total} P}$$

$$P = 2\pi r_2 = 0.127\,m$$

$$h_{total} = \frac{1}{R'_{out} P}$$

$$h_{total} \approx 11.3\,W/m^2K$$

9. Radiation Contribution

$$T_s \approx 49°C = 322\,K$$

$$T_\infty = 304\,K$$

$$h_r = \sigma\epsilon(T_s + T_\infty)(T_s^2 + T_\infty^2)$$

$$h_r \approx 6.5\,W/m^2K$$

10. Final Convection Coefficient

$$h_o = h_{total} - h_r$$

$$h_o \approx 4.8\, W/m^2K$$

11. Conclusion

The system is modeled using an energy balance with thermal resistance to account for radial heat transfer through insulation and convection-radiation at the surface. The obtained value of h_o lies in the expected range for natural convection in air.